

Average inductor current sensor for digitally-controlled switched-mode power supplies

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Abstract—Current-mode control in digitally-controlled switched-mode power supplies typically requires analog-to-digital (A/D) conversion of at least two signals, voltage and current. The complexity of voltage A/D converters can be reduced using window A/D techniques. In conventional current A/D conversion, however, relatively high resolution is required over a wide range of signals, which results in increased complexity, power consumption and cost of the controller. This paper proposes a very simple feedback sensor capable of high-resolution average inductor current sensing using two analog comparators and an analog low-pass filter. The approach requires very few external components and employs minimal digital hardware resources. A dynamic model and performance of the average inductor current sensor are experimentally verified on a 12 V input, 19 V output, 50 W boost converter prototype.

I. INTRODUCTION

In the area of digital control of high-frequency Switched-Mode Power Supplies (SMPS), numerous advances have been made in terms of reduced controller complexity or improved performance, together with enhanced programmability and flexibility features. Practical digital voltage-mode control, requiring relatively small hardware resources, has been demonstrated in a number of applications [1]. Techniques such as window-flash analog-to-digital (A/D) conversion have been applied to reduce the controller complexity [2]. Digital current-mode control, however, requires two A/D conversions, as shown in the boost converter example of Fig. 1. Furthermore, relatively high-resolution current A/D conversion is required over a wide range of sensed currents, which increases complexity, power consumption and cost of the controller implementation.

Several alternatives to address this issue have been proposed: for instance, specific low-resolution A/D structures for current sensing are used in [3] to implement average current mode control. The use of an external comparator and digital-to-analog (D/A) converter that can be easily implemented in a digital system has recently been proposed: [4] uses such a scheme, embedded in a feedback loop, to estimate the instantaneous inductance current, while [5] proposes a simpler system suitable to implement peak current mode control.

This paper proposes a simple high-resolution average inductor current sensor for digitally-controlled SMPS. It is inspired by [4], and by the A/D concept proposed in [6] and

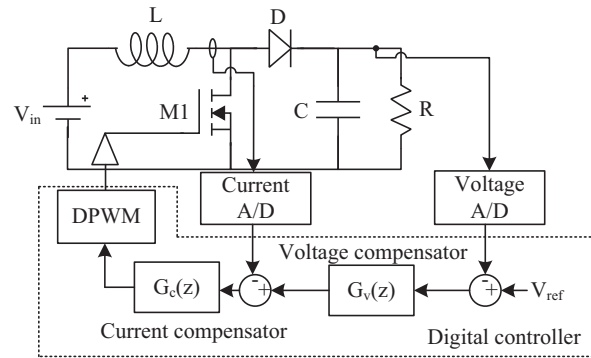


Figure 1. Digitally-controlled boost converter with current-mode control.

[7], where the output voltage ripple was used to indirectly measure the output voltage using a single analog comparator. The proposed sensor takes advantage of the inductor current waveform to measure its average value using very few external components: two comparators and a simple analog low pass filter are required. It can be easily implemented in digital logic (FPGA or custom chip) using few hardware resources, and it provides accurate sensing of the average inductor current in both continuous conduction mode (CCM) and discontinuous conduction mode (DCM).

This paper is organized as follows: operation of the sensor is described in Section II. A dynamic model is derived in Section III. Section IV presents results of simulation of the proposed sensor in open and closed loop conditions. Section V shows experimental results obtained with a 12 V input, 19 V output, 50 W boost converter. Finally, conclusions are given in Section VI.

II. OPERATION OF THE AVERAGE INDUCTOR CURRENT SENSOR

Figure 2 shows the proposed sensor architecture, while Fig. 3a shows the operating waveforms at the inputs of the main comparator. From Fig. 3a and using simple geometrical considerations, the following equation can be obtained:

$$d_p = \frac{t_p}{T_{sw}} = d_{ref} + g_{comp} ((v_+) - v_-), \quad (1)$$

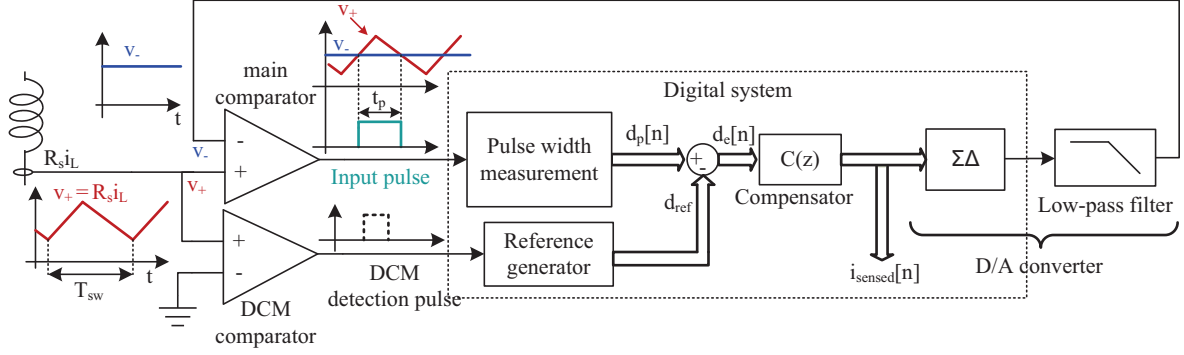


Figure 2. Proposed current sensor.

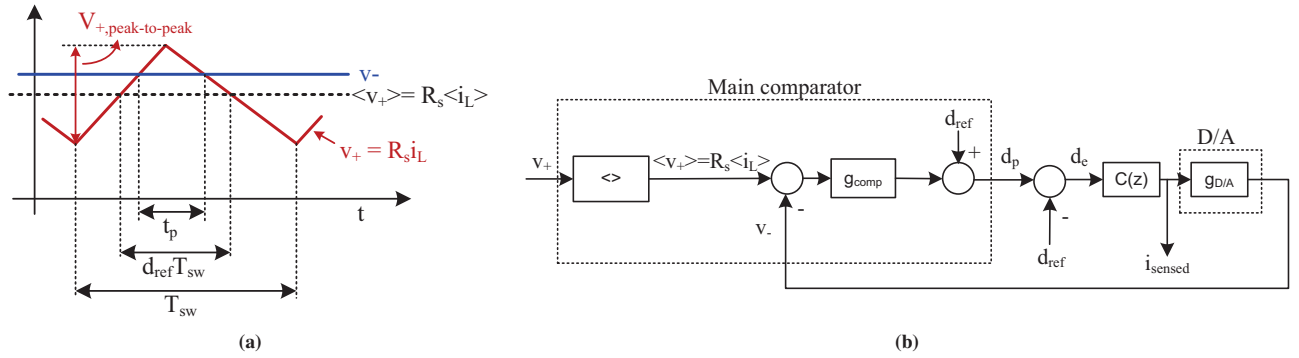


Figure 3. (a) Operating waveforms of the proposed current sensor; (b) block diagram of the system.

d_p being the duty cycle of the pulse obtained at the output of the comparator, g_{comp} the equivalent gain of the comparator and d_{ref} a value that depends on the conduction mode of the inductor. Note from (1) that d_{ref} equals the value of d_p when $\langle v_+ \rangle = v_-$: this fact will be used to deduce d_{ref} in Sections II-A and II-B. The notation $\langle \cdot \rangle$ indicates the averaging operation over one switching cycle. g_{comp} can be easily deduced taking into account that the operation of the comparator is similar to that of a conventional pulse-width modulator,

$$g_{comp} \approx G_{comp} = \frac{1}{V_{+,peak-to-peak}}, \quad (2)$$

G_{comp} being the value of g_{comp} when the inherent one-cycle delay of the comparator operation is neglected, and $V_{+,peak-to-peak}$ the amplitude of the triangular waveform input (as shown in Fig. 3a), which in turn depends on the inductor current ripple. The pulse width measurement block turns the output pulse duty cycle, d_p , into the digital signal $d_p[n]$. After that, the quantity d_{ref} is subtracted from $d_p[n]$, thus generating an error signal $d_e[n]$ which is the input to the sensor loop compensator $C(z)$. The output of the compensator, $i_{sensed}[n]$, is turned back to the analog domain by a D/A converter formed by a 2nd-order $\Sigma - \Delta$ modulator and a simple, first-order low-pass filter. The filter output is finally returned to the inverting input of the comparator.

Taking into account (1) and Fig. 2, the block diagram shown in Fig. 3b can be obtained, where $g_{D/A}$ is the transfer function of the D/A conversion stage. From this diagram, the transfer function between i_{sensed} and $\langle i_L \rangle$ is:

$$\frac{i_{sensed}}{\langle i_L \rangle} = R_s \frac{T_{sensor}}{1 + T_{sensor} \cdot g_{D/A}}, \quad (3)$$

T_{sensor} being defined as

$$T_{sensor} = g_{comp} C(z). \quad (4)$$

If the loop gain, $T_{sensor} g_{D/A}$ fulfills $\|T_{sensor} g_{D/A}\| \gg 1$, then (3) becomes:

$$i_{sensed} = \frac{R_s}{g_{D/A}} \langle i_L \rangle. \quad (5)$$

Neglecting delays, $g_{D/A}$ can be approximated by $G_{D/A}$, and thus (5) can be written as:

$$i_{sensed} \approx \frac{R_s}{G_{D/A}} \langle i_L \rangle. \quad (6)$$

Therefore, the internal signal i_{sensed} is a digital representation of the average inductor current. The following sections establish the appropriate value for the reference pulse width, d_{ref} , to ensure that the average inductor current can be precisely

measured in both DCM and CCM.

A. Continuous Conduction Mode operation

The waveforms in Fig. 3a correspond to Continuous Conduction Mode (CCM) operation. Simple geometrical considerations imply that $d_{ref} = 0.5$, i.e. when $\langle v_+ \rangle = v_-$ the width of the output pulse of the comparator equals 0.5. Thereby, in CCM operation (1) can be expressed as:

$$d_p = \frac{t_p}{T_{sw}} = 0.5 + g_{comp} (\langle v_+ \rangle - v_-), \quad (7)$$

and a pulse width equal to 0.5 is used as the reference in the sensor feedback loop to achieve the desired operation.

B. Discontinuous Conduction Mode operation

Figure 4 shows the operating waveforms when the inductor is operating in Discontinuous Conduction Mode (DCM). Once again, simple geometrical considerations on Fig. 4 lead to the following expression for d_{ref} :

$$d_{ref} = (D + \Delta) \cdot \left(1 - \frac{D + \Delta}{2}\right). \quad (8)$$

Equation (8) can easily be extended to CCM operation by substituting $\Delta = 1 - D$. Measurement of the quantity $D + \Delta$ requires a detection of the zero-crossing instant of the inductor current, which is the function of the DCM comparator shown in Fig. 2.

This analysis demonstrates that a simple adjustment of the reference signal of the sensor feedback loop allows an appropriate measurement of the average inductor current regardless of the conduction mode.

III. DYNAMIC MODEL

As the proposed sensor operates in closed loop, a dynamic analysis is required to ensure its stability and dynamic performance during transients. Figure 5 shows a small-signal dynamic model of the proposed sensor, which can be easily obtained from Fig. 3b. The perturbed variables are indicated with hats. The dynamic analysis performed in this section is valid only for CCM operation, as additional effects caused by the variation of d_{ref} that takes place in DCM operation have been neglected to simplify the analysis. However, the same

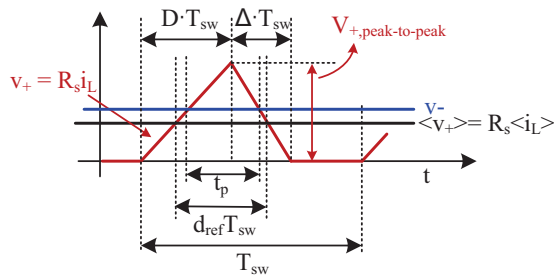


Figure 4. Operating waveforms of the proposed current sensor in DCM.

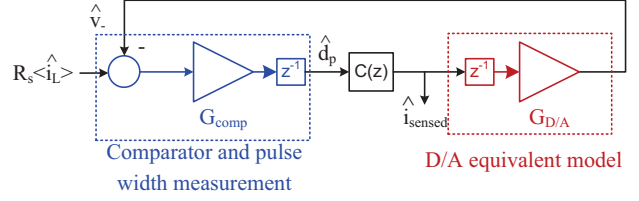


Figure 5. Dynamic model of the proposed sensor.

methodology can easily be applied in DCM operation. Note that the sampling period of the Z-domain transfer functions is equal to the switching period.

The complete transfer function for the comparator stage can be expressed as:

$$g_{comp}(z) = G_{comp} z^{-1}, \quad (9)$$

G_{comp} being the value used in Section II for g_{comp} when the delay was neglected. As G_{comp} depends on the current slope, its value changes with the converter topology. For instance, in the case of the boost converter, (9) becomes:

$$g_{comp}(z) = \frac{L}{R_s D T_{sw} V_{in}} z^{-1}, \quad (10)$$

D being the steady-state duty cycle, V_{in} being the converter input voltage, L the inductance value and R_s the sensor gain (in volts per amp). In the case of the buck converter, (9) becomes:

$$g_{comp}(z) = \frac{L}{R_s (1 - D) T_{sw} V_{out}} z^{-1}, \quad (11)$$

V_{out} being the output voltage. Note that in both cases G_{comp} is expressed in V^{-1} , as expected from (2). In general, the comparator gain can be expressed as:

$$g_{comp}(z) = \frac{1}{R_s \Delta i_L} z^{-1}, \quad (12)$$

Δi_L being the inductor current ripple.

As the $\Sigma - \Delta$ modulator inside the D/A operates at a much higher frequency than the rest of the loop, the dynamic behavior of the low-pass filter can be neglected. Thereby, a simple Zero Order Hold (ZOH) model with gain can be used for the D/A:

$$g_{D/A}(z) = G_{D/A} z^{-1}. \quad (13)$$

The constant $G_{D/A}$ depends on the D/A implementation, i.e. on the amplitude of the digital signal at the output of the $\Sigma - \Delta$ modulator.

The simplest $C(z)$ that guarantees zero steady-state error is an integrator,

$$C(z) = \frac{K}{1 - z^{-1}}. \quad (14)$$

Substituting (10), (13) and (14) into (3):

$$H_{\text{sensor}}(z) = \frac{\hat{i}_{\text{sensed}}}{\langle i_L \rangle} = \frac{R_s z G_{\text{comp}} K}{(z^2 - z + G_{\text{comp}} G_{D/A} K)} \quad (15)$$

The closed loop poles of (15) can be used to derive stability conditions and to analyze the theoretical dynamic performance of the proposed sensor. For instance, in the case of the boost converter, stability is guaranteed if:

$$K < K_{\text{limit,boost}} = \frac{1}{G_{\text{comp}} G_{D/A}} = \frac{R_s D T_{\text{sw}} V_{\text{in}}}{L G_{D/A}} \quad (16)$$

while in the case of the buck converter the maximum K value is:

$$K < K_{\text{limit,buck}} = \frac{1}{G_{\text{comp}} G_{D/A}} = \frac{R_s (1-D) T_{\text{sw}} V_{\text{out}}}{L G_{D/A}} \quad (17)$$

Once again, in general the stability condition can be expressed as

$$K < K_{\text{limit}} = \frac{R_s \Delta i_L}{G_{D/A}} \quad (18)$$

Equation (18) relates the dynamic performance of the sensor with the design of the power stage; the inductor current ripple is the only power stage design parameter that determines the achievable bandwidth of the proposed sensor. Therefore, for a certain inductor current ripple, the dynamic performance achievable is independent of the switching frequency, duty cycle or input or output voltages.

A. Buck converter example

In this section, the proposed current sensor was designed to measure the inductor current of a buck converter with the following parameters: $V_{\text{in}} = 12$ V, $V_{\text{out}} = 3.3$ V, $f_{\text{sw}} = 250$ KHz, $L = 5.6$ μ H, $C = 15$ μ F. The rated load was $R_{\text{load}} = 0.33$ Ω . The rest of the parameters were $G_{D/A} = 3.3$ V/A and $R_s = 0.2$ Ω . It was assumed that the final goal was to design an inner current loop: to do so, the loop gain, comprised as usual by the power stage plant and the transfer function of the sensor, was analyzed.

The small-signal transfer function from the duty cycle to the inductor current, $G_{iL,d}$, and the transfer function of the sensor, H_{sensor} , given by (15), are plotted in Fig. 6. It can be appreciated that the cut-off frequency of H_{sensor} is beyond 10 KHz, and therefore a relatively high bandwidth can be achieved in the current loop. However, compared to a conventional analog average-current control design (in which H_{sensor} is constant), it can be seen that the proposed sensor adds high frequency poles to the loop gain, thus complicating the design of the current loop compensator and, in practice, limiting the achievable bandwidth.

B. Boost converter example

The proposed current sensor was designed to measure the input current in a boost converter with the following parameters: $V_{\text{in}} = 12$ V, $V_{\text{out}} = 19$ V, $f_{\text{sw}} = 100$ KHz, $L = 35$ μ H,

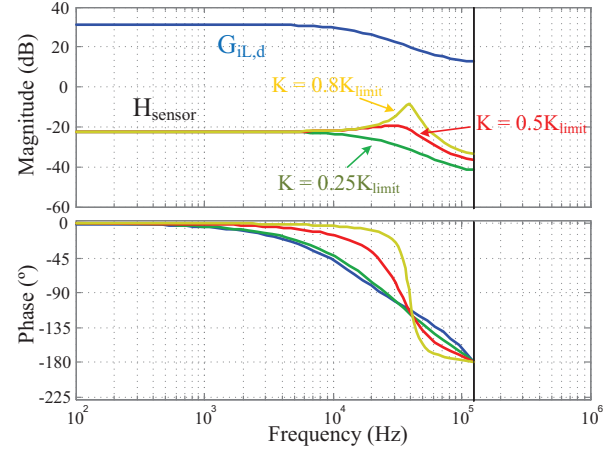


Figure 6. Bode diagram of the most representative transfer functions in the Buck converter design: $G_{iL,d}$ and H_{sensor} are shown, the latter for different values of K .

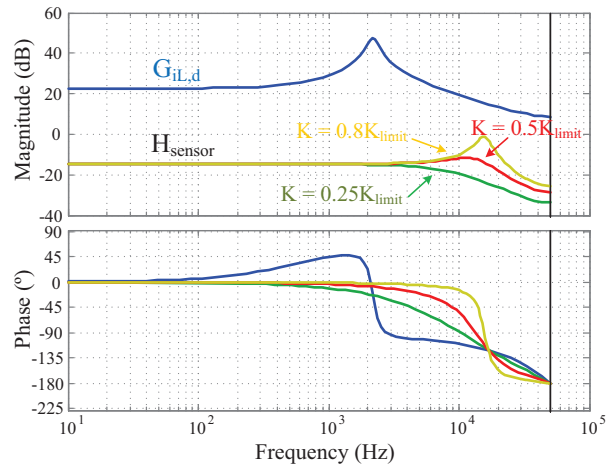


Figure 7. Bode diagram of the most representative transfer functions in the boost converter design: $G_{iL,d}$ and H_{sensor} are shown, the latter for different values of K .

$C = 60$ μ F. The rated load was $R_{\text{load}} = 8$ Ω . The rest of the parameters were $G_{D/A} = 3.3$ and $R_s = 0.605$ Ω . The small-signal transfer function from the duty cycle to the inductor current, $G_{iL,d}$, and the transfer function of the sensor given by (15) are plotted in Fig. 7. Once again, the sensor achieves a relatively high bandwidth but, as in the case of the buck converter, the additional high frequency poles add extra phase lag, therefore limiting the dynamic response obtainable from the current loop.

Compared to a traditional analog average current-mode control [9], the averaging operation performed by the sensor includes additional high frequency poles in the loop gain that have to be taken into account when designing the current loop, and that might limit the achievable dynamic performance.

IV. SIMULATION RESULTS

A boost converter and the proposed sensor were simulated in Simulink. The power stage reactive components were: $L = 35 \mu H$, $C = 60 \mu F$ and the switching frequency was $f_{sw} = 100$ kHz. The input and output voltages were $V_{in} = 12$ V and $V_{out} = 19$ V, respectively, while the rated load was $R_{load} = 8 \Omega$. Table I shows the remaining design parameters. The D/A was simulated using a 2nd-order $\Sigma - \Delta$ [8] operating at 100 MHz, along with a first order low-pass filter with a cut-off frequency close to 200 kHz.

The current sensor compensator, $C(z)$, was chosen as in (14), and after that Sisotool was used to find a value of K that yielded an appropriate phase margin and closed-loop bandwidth. A design with approximately $K = 0.45K_{limit}$ was chosen, yielding a phase margin of 52° and bandwidth close to 10 kHz for the current sensor.

A. Open loop simulations

Figs. 8a and 8b show the sensor in steady-state. Fig. 8a shows the estimated digital current, i_{sensed} , once scaled, along with the actual inductor current, i_L , and the average inductor current $\langle i_L \rangle$ obtained by low-pass filtering i_L using a second order filter. It can be seen that the scaled version of i_{sensed} precisely matches the average inductor current. Fig. 8b shows the same waveforms depicted in Fig. 8a but when the inductor operates in DCM: it can be seen that the sensor still provides the actual average current, therefore confirming the validity of the approach described in Section II-B. As already mentioned, the required value of d_{ref} given by (8) was obtained by measuring the inductor conduction period: according to Fig. 4, the duration of such interval directly provides the quantity $(D + \Delta)T_{sw}$.

Figs. 8c and 8d show the sensor response under a load step. In spite of the noticeably underdamped response shown by the inductor current, the sensor keeps track of the average current. Fig. 8d shows the response in detail, where it can be observed the delay that takes place in the current measurement process.

B. Closed loop simulations

As an example, the proposed current sensor was used to control the input current of the same boost converter. Fig. 9 shows an overall view of the system. The current compensator, G_c , was designed using Sisotool: a simple integrator with gain was used to simplify the system. Note that, with this simple compensator and taking into account the actual shape of $G_{iL,d}$ shown in Fig. 7, the system will have a relatively underdamped response and a settling time in the milliseconds

Table I. Main parameters of the simulation and experimental results presented in this section.

G_{comp} ($\frac{1}{V}$)	$G_{D/A}$ ($\frac{V}{A}$)	R_s ($\frac{V}{A}$)	K (A)	K_{limit} (A)
0.729	3.3	0.605	0.1	0.23

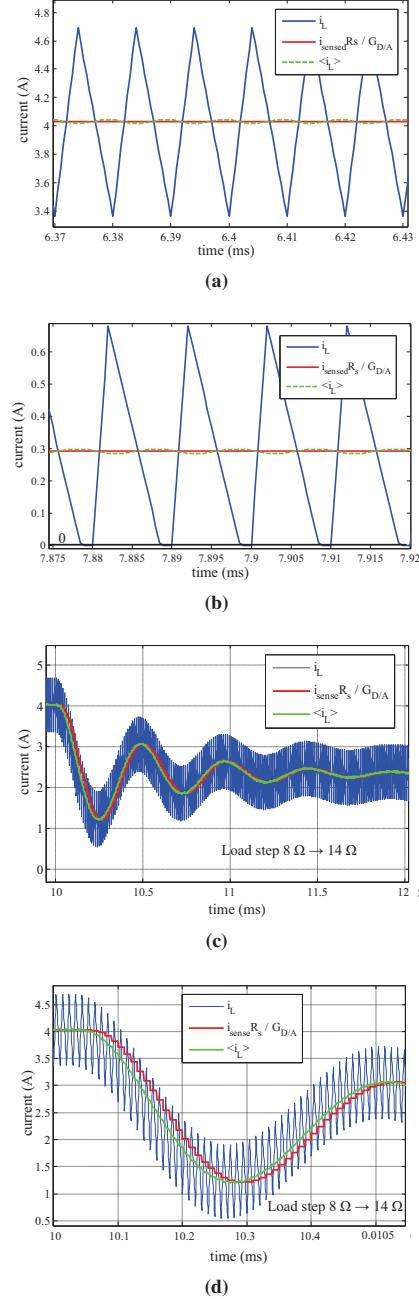


Figure 8. Simulation results: (a) CCM steady-state operation (b) DCM steady-state operation; (c) 100 % to 50 % load step; (d) detailed view of the waveform in 8c.

range. Therefore, it is worth to remark that the goal was only to test the feasibility of the system in closed loop operation.

Figure 10a shows the response of the system when a step was applied to the current reference, I_{ref} . Once again, a noticeable underdamped response can be appreciated. The input current follows the reference command and the sensor also tracks the inductor current, as desired. Figure 10b shows a load step, while the reference current is kept constant. The system undergoes a transient and, after a few milliseconds,

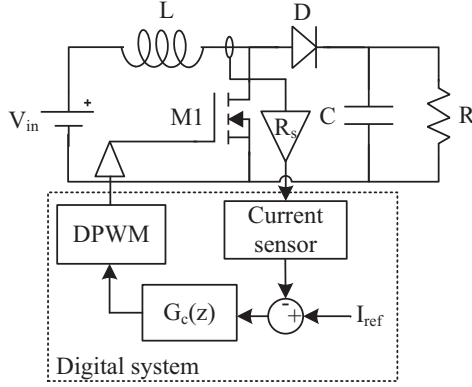


Figure 9. Schematic diagram of the system in closed loop operation.

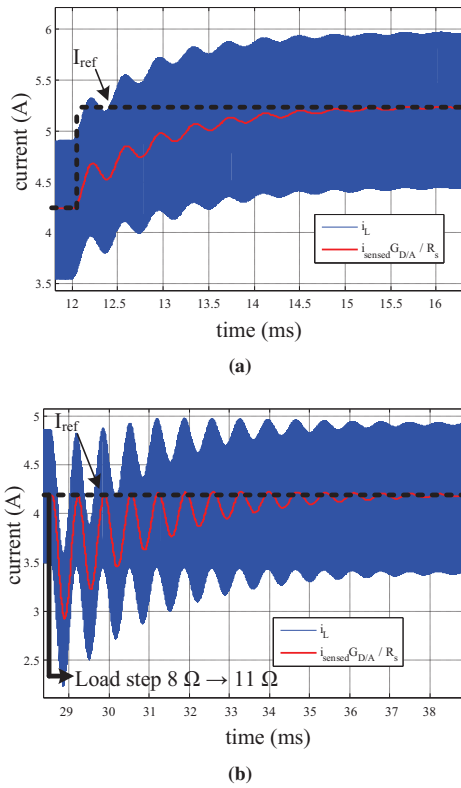


Figure 10. Simulated inductance current and sensed current (scaled) in closed loop operation: (a) step in the reference current; (b) load step.

the current loop forces the input current to return to its original value. Both simulations show that the overall system (sensor plus converter) is capable of operating in a closed loop manner.

V. EXPERIMENTAL RESULTS

A boost converter with the same parameters described in Section IV was built and tested. This 12 V input, 19 V output, 50 W converter, designed to supply laptop computers from a 12 V battery, was used to test the proposed current sensor.

Fig. 11a shows the experimental setup: an additional hardware board with the analog current sense circuitry was attached to the main board. The input current was sensed using a 5 mΩ sense resistor in series with the inductance, while an INA138 current shunt monitor from Texas Instruments translated the differential voltage across the sense resistor to a ground-referenced signal. Then, a dual OPA2727 was used to amplify the signal provided by the INA138 in two stages, each of them with a gain equal to 10. The theoretical overall sense gain, R_s , was thus 0.5; due to component tolerances, the actual gain measured in the circuit was $R_s = 0.605$. Two low cost, precision comparators (TL3016 from Texas Instruments) were required: both are shown in Fig. 2. Finally, the board included the first order low-pass filter of the D/A converter. Figure 11b shows an schematic diagram of the converter power stage, along with the sense circuitry.

The sensor was implemented using a Virtex 4 FPGA Development Board, and all the digital hardware was programmed in VHDL. The value of K was set close to $0.45K_{limit}$, to match as much as possible the simulation scenario described in Section IV. The remaining required digital hardware, including the digital pulse width modulator used to control the power MOSFET or the current compensator for closed loop operation, were also implemented in the FPGA.

A. Steady-state measurements

Figure 12 shows steady-state measurements that were obtained to test the accuracy of the sensor. The average input

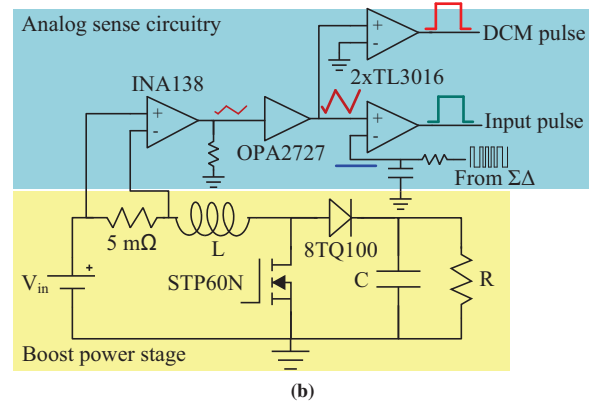
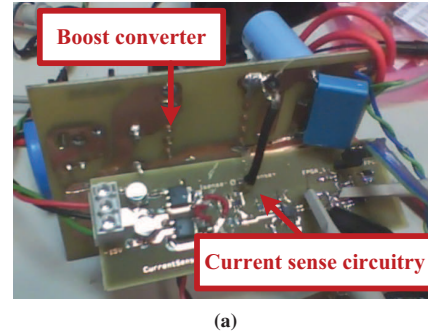


Figure 11. (a) Experimental setup; (b) schematic diagram of the power stage and the analog current sense circuitry.

current was measured using a precision multimeter, along with an HP current shunt, model 34330A. The internal resolution of the sensor, i.e. the resolution chosen for i_{sensed} , was set to 10 bits. The worst-case error measured was less than 2 %. Thus, Fig. 12 demonstrates that the sensor is capable of accurately measuring the average input current.

Figs. 13a and 13b show both inputs of the comparator, signals $v_+ = R_s i_L$ and v_- , the analog scaled version of i_{sensed} , when the converter is in steady-state and for two different load values. It can be seen that the average value of both signals is the same, as was predicted by (6), and that the sensor performs as expected in steady-state.

Fig. 14a shows the waveforms when the converter operates in DCM and $d_{ref} = 0.5$, i.e. the sensor was designed only for CCM operation. The actual average input current was close to 1.48 A, while the current measured by the sensor was 1.1 A, i.e. the relative error was close to 25 %. Fig. 14b shows the operation of the system when d_{ref} was appropriately modified

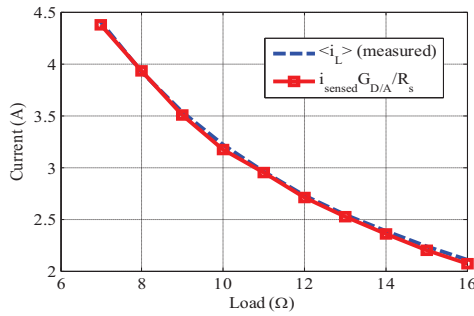


Figure 12. Steady-state measured average input current, $\langle i_L \rangle$, and v_- (the analog scaled version of i_{sensed}).

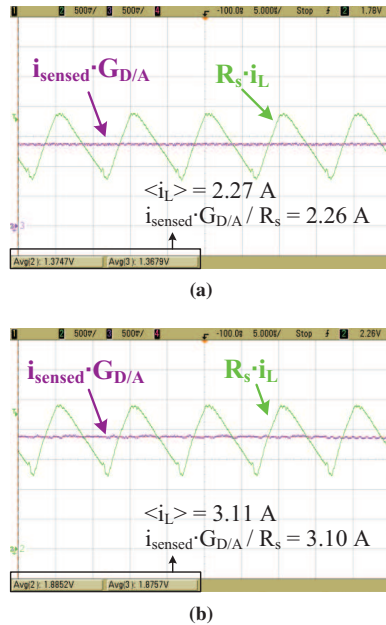


Figure 13. Experimental results: (a) steady-state operation (14 Ω load); (b) steady-state operation (10 Ω load);

according to (8). The actual input current was then 1.49 A, and the measured current was 1.514 A, i.e. the error dropped to 1.6 %, a value consistent with the results shown in Fig. 12. Note that, to obtain the results shown in Figs. 14a and 14b, not only a lighter load was used but also the switching frequency was divided by two; this was done to highlight the relative measurement error obtained when the sensor was programmed only for CCM operation. With a higher switching frequency the relative error of 25 % would have been noticeably smaller.

B. Dynamic performance

Several experiments were carried out to test the dynamic performance of the system. First, load steps were programmed with the converter operating in an open loop manner. Fig. 15 shows the positive input of the comparator, $v_+ = i_L R_s$, and the scaled version of the sensed current, $v_- = i_{sensed} G_{D/A}$, under 8 Ω to 14 Ω and 14 Ω to 8 Ω load steps, respectively. It can be seen that the sensor tracks the average inductor current, as expected. Furthermore, the results resemble those obtained in simulation in Section IV. However, higher damping and smaller settling time can be appreciated, probably caused by non-idealities that were not appropriately accounted for in the simulations, e.g. the actual MOSFET and inductor parasitic resistances, and also by a slightly smaller inductance value in the experimental prototype.

The closed loop scheme described in Section IV and depicted in Fig. 9 was also tested experimentally. Once again, the goal was to demonstrate the feasibility of closed loop operation in practice. The same current compensator used in Section IV was implemented in the FPGA. Fig. 16a shows a step in the reference current of the loop, I_{ref} . It can be appreciated that the average input current follows the reference, as expected. Note that the results resemble those

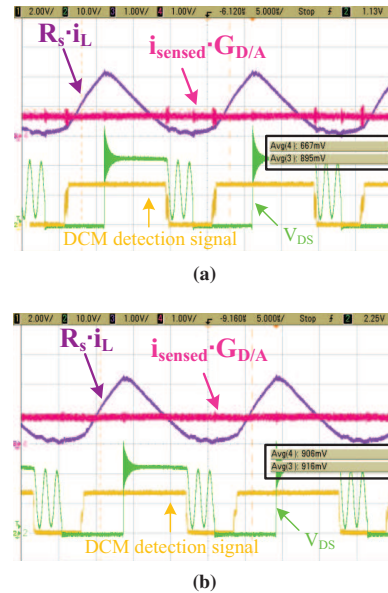


Figure 14. Experimental results in DCM operation: (a) d_{ref} is kept constant and equal to 0.5; (b) d_{ref} is modified according to (8).

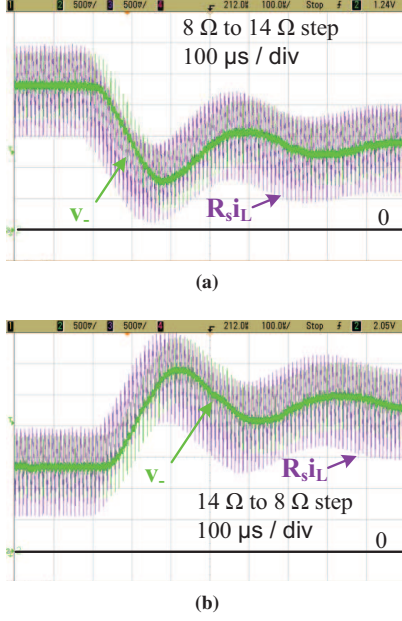


Figure 15. (a) 8 Ω to 14 Ω; (b) 14 Ω to 8 Ω.

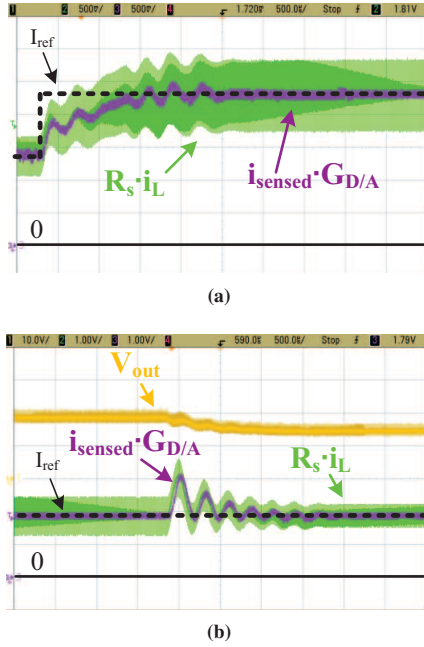


Figure 16. Experimental inductance current and sensed current (scaled) in closed loop operation: (a) step in the reference current; (b) 8 Ω to 11 Ω load step.

obtained in Fig. 10a, except for a higher damping and a shorter settling time. Fig. 16b shows a load step (8 to 11 Ω) while the reference current was kept constant: the system underwent a transient but recovered after a few milliseconds and went back to its original state. Therefore, it is apparent that the system was able to operate in closed loop operation.

VI. CONCLUSIONS

An average inductor current sensor for digitally-controlled switched mode power supplies has been presented in this paper. The proposed sensor allows the average inductor current to be precisely measured without a complex A/D converter. As it operates in a closed loop manner, its dynamic performance and stability conditions were analyzed in this work. It was shown that the proposed sensor was capable of reaching relatively high bandwidths. Its influence over the loop gain of the converter, by means of additional high frequency poles that might limit the achievable dynamic performance compared to conventional analog designs, was also highlighted.

Thorough simulations were presented to demonstrate the feasibility of the sensor concept, and experimental results were obtained by means of a boost converter prototype. Both open and closed loop operation were tested.

The proposed sensor can help simplify hardware implementation of digitally controlled current-mode SMPS, and seems especially suitable for Power Factor Correction applications, where no stringent bandwidth requirements are needed. However, it has been demonstrated in this paper that it can be generally applied to any kind of average inductor current measurement. Further developments might deal with real time tuning of the sensor gain to optimize the response for different operating conditions of the converter, as well as a deeper study of the influence of the sensor on the loop gain, including non-linearities and limit cycling phenomena.

ACKNOWLEDGEMENT

This work was supported by the Spanish Ministry of Science and Education under grant AP2006-04777 and project TEC-2007-66917, and by the EU under the FEDER program.

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